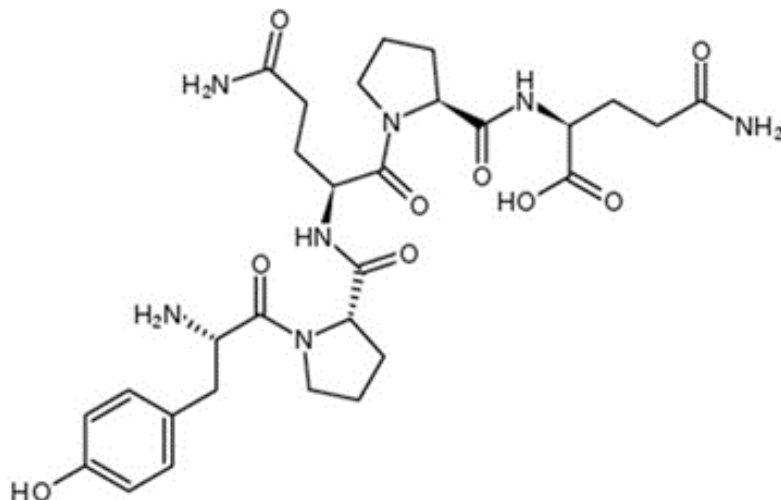


****These questions are basic questions to get you started with your studying.****

Chapter 4 Practice Problems

1. Which of the following function groups will make a molecule hydrophilic?
 - a. Sulfhydryl
 - b. Methyl
 - c. Hydroxyl
 - d. Amino
2. Which of the following functional groups is important for energy?
 - a. Carboxyl
 - b. Hydroxyl
 - c. Phosphate
 - d. Sulfhydryl
3. What differs among isomers of the same molecule?
 - a. Number of protons
 - b. Number of neutrons
 - c. The arrangement of the molecules
 - d. The molecular formula
4. In a carbonyl group, _____ refers to carbon double bonded to an oxygen, occurring at the end of a chain
 - a. Aldehyde
 - b. Ketone
5. How can carbon chains vary?
 - a. By a single, double, or triple bond
 - b. Ring structures
 - c. Length
 - d. Carbon chains can be branched or linear
 - e. All of the above
6. Hydrocarbons are _____
 - a. Hydrophobic
 - b. Hydrophilic
7. Which of the following types of isomers are mirror images of each other?
 - a. Structural
 - b. Geometric
 - c. Enantiomer
8. Isomers can be expected to...
 - a. Have the same molecular formula

- b. Have different properties
- c. Have a different 3D representation
- d. All of the above



9. List the functional groups in the above molecule.
 - a. Amine, aldehyde
 - b. Methyl, ketone, sulfhydryl
 - c. Hydroxyl, ketone, amine, carboxyl
 - d. Aldehyde, ketone, methyl
10. Describe the difference between structural isomers, cis-trans geometric isomers, and enantiomers
 1. C
 2. C
 3. C
 4. A
 5. E
 6. A
 7. C
 8. D
 9. C
10. Isomers have the same molecular formula. Structural isomers are different arrangements of covalent bonds. Enantiomers are mirror images of each other. Cis-trans isomers have different arrangements around a double bond.